

Newman-Penrose Formalism Calculations

Stephani et al., Equation (36.34)

Metric

$$ds^2 = -F^2 dt \otimes dt + dr \otimes dr + d\vartheta \otimes d\vartheta + \left(\frac{\partial F}{\partial \vartheta} \right)^2 d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, r, \vartheta, \phi)$$

Metric Functions

$$F_{,\vartheta\vartheta} = \frac{b}{F}$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}} F \right) dt + \left(\sqrt{\frac{1}{2}} \right) d\vartheta$$

$$n = \left(\sqrt{\frac{1}{2}} F \right) dt + \left(-\sqrt{\frac{1}{2}} \right) d\vartheta$$

$$m = \left(\sqrt{\frac{1}{2}} \right) dr + \left(i \sqrt{\frac{1}{2}} \frac{\partial}{\partial \vartheta} F \right) d\phi$$

$$\overline{m} = \left(\sqrt{\frac{1}{2}} \right) dr + \left(-i \sqrt{\frac{1}{2}} \frac{\partial}{\partial \vartheta} F \right) d\phi$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2F} \right) dt + \left(\sqrt{\frac{1}{2}} \right) d\vartheta$$

$$n = \left(-\frac{\sqrt{2}}{2F} \right) dt + \left(-\sqrt{\frac{1}{2}} \right) d\vartheta$$

$$m = \left(\sqrt{\frac{1}{2}} \right) dr + \left(\frac{i \sqrt{2}}{2 \frac{\partial}{\partial \vartheta} F} \right) d\phi$$

$$\overline{m} = \left(\sqrt{\frac{1}{2}} \right) dr + \left(-\frac{i \sqrt{2}}{2 \frac{\partial}{\partial \vartheta} F} \right) d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2F} \partial_t + \sqrt{\frac{1}{2}} \partial_\vartheta$$

$$\Delta = -\frac{\sqrt{2}}{2F} \partial_t - \sqrt{\frac{1}{2}} \partial_\vartheta$$

$$\delta = \sqrt{\frac{1}{2}} \partial_r + \frac{i \sqrt{2}}{2 \frac{\partial}{\partial \vartheta} F} \partial_\phi$$

$$\bar{\delta} = \sqrt{\frac{1}{2}} \partial_r - \frac{i \sqrt{2}}{2 \frac{\partial}{\partial \vartheta} F} \partial_\phi$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2}\frac{\partial^2}{(\partial\vartheta)^2}F}{4\frac{\partial}{\partial\vartheta}F}$$

$$\sigma = \frac{\sqrt{2}\frac{\partial^2}{(\partial\vartheta)^2}F}{4\frac{\partial}{\partial\vartheta}F}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}\frac{\partial^2}{(\partial\vartheta)^2}F}{4\frac{\partial}{\partial\vartheta}F}$$

$$\lambda = \frac{\sqrt{2}\frac{\partial^2}{(\partial\vartheta)^2}F}{4\frac{\partial}{\partial\vartheta}F}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{\sqrt{2}\frac{\partial}{\partial\vartheta}F}{4F}$$

$$\gamma = \frac{\sqrt{2} \frac{\partial}{\partial \vartheta} F}{4 F}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{\frac{\partial^2}{(\partial \vartheta)^2} F}{4 F} - \frac{\frac{\partial^3}{(\partial \vartheta)^3} F}{4 \frac{\partial}{\partial \vartheta} F}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = \frac{\frac{\partial^2}{(\partial \vartheta)^2} F}{4 F} + \frac{\frac{\partial^3}{(\partial \vartheta)^3} F}{4 \frac{\partial}{\partial \vartheta} F}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{\frac{\partial^2}{(\partial \vartheta)^2} F}{4 F}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = \frac{\frac{\partial^2}{(\partial \vartheta)^2} F}{4 F} + \frac{\frac{\partial^3}{(\partial \vartheta)^3} F}{4 \frac{\partial}{\partial \vartheta} F}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{\frac{\partial^2}{(\partial \vartheta)^2} F}{4 F} - \frac{\frac{\partial^3}{(\partial \vartheta)^3} F}{4 \frac{\partial}{\partial \vartheta} F}$$

$$\Lambda = -\frac{\frac{\partial^2}{(\partial\vartheta)^2}F}{6\,F} - \frac{\frac{\partial^3}{(\partial\vartheta)^3}F}{12\,\frac{\partial}{\partial\vartheta}F}$$

Weyl Tensor Components

$$\Psi_0 = -\frac{\frac{\partial^2}{(\partial\vartheta)^2}F}{4\,F} + \frac{\frac{\partial^3}{(\partial\vartheta)^3}F}{4\,\frac{\partial}{\partial\vartheta}F}$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{\frac{\partial^2}{(\partial\vartheta)^2}F}{12\,F} - \frac{\frac{\partial^3}{(\partial\vartheta)^3}F}{12\,\frac{\partial}{\partial\vartheta}F}$$

$$\Psi_3 = 0$$

$$\Psi_4 = -\frac{\frac{\partial^2}{(\partial\vartheta)^2}F}{4\,F} + \frac{\frac{\partial^3}{(\partial\vartheta)^3}F}{4\,\frac{\partial}{\partial\vartheta}F}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"